

Network Threats and Mitigation & Physical Security and Risks

Task 1: Create a table listing 5 common network threats (such as phishing, DDoS, man-in-the-middle, malware, and password attacks), and for each, write one real-world example from an app or service you use (like WhatsApp, Instagram, or Paytm).

Threat	Real-World Example
Phishing	Fake WhatsApp/Paytm messages claiming "your KYC will be blocked" with a link to a lookalike login page that steals credentials
DDoS (Denial of Service)	Large platforms like Instagram occasionally go down for users worldwide when attackers flood their servers with junk traffic, overwhelming the infrastructure
Man-in-the-Middle (MITM)	Someone on the same public/café WiFi intercepting an unencrypted Paytm or banking session to capture login data between your phone and the server
Malware	Modded WhatsApp APKs (like "GB WhatsApp") downloaded outside the Play Store that bundle spyware or trojans along with the extra features
Password Attacks	Credential stuffing on Instagram — attackers reuse leaked username/password combos from other breached sites to break into accounts where people reused passwords

Task 2 :Simulate a brute-force attack scenario by writing a simple Python script that tries to guess a 4-digit PIN code by checking all possible combinations against a preset value. Constraint: The script should print the number of attempts it took to guess the correct PIN.

```
import itertools
import time

def brute_force_pin(actual_pin):
    attempts = 0
    start_time = time.time()
```

```
# Try every possible 4-digit combination from 0000 to 9999
for a, b, c, d in itertools.product(range(10), repeat=4):
    guess = f"{a}{b}{c}{d}"
    attempts += 1

    if guess == actual_pin:
        elapsed = time.time() - start_time
        print(f"PIN cracked: {guess}")
        print(f"Total attempts taken: {attempts}")
        print(f"Time taken: {elapsed:.4f} seconds")
        return attempts

print("PIN not found (shouldn't happen for a valid 4-digit PIN)")
return attempts

if __name__ == "__main__":
    preset_pin = "7842" # the PIN being "attacked" — change this to test
    brute_force_pin(preset_pin)
```

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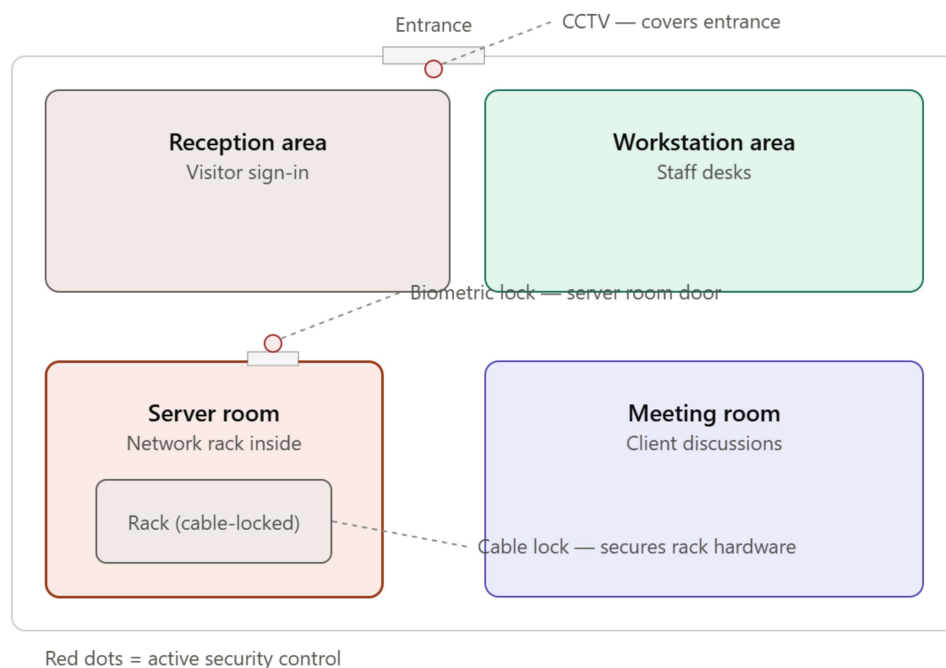
... PIN cracked: 7842
Total attempts taken: 7843
Time taken: 0.0057 seconds
```

Task 3 :List 4 physical security risks that could affect a server room hosting a popular app like Zomato or BookMyShow, and suggest one mitigation step for each risk.

Risk		Mitigation
Unauthorized access/intrusion	physical	Biometric or smart-card access control on the server room door, logged and restricted to authorized staff only
Fire damage		Install smoke detectors + a clean-agent (non-water) fire suppression system, since water-based sprinklers would destroy the hardware
Power outage / electrical surge		UPS backup + surge protectors + a generator for extended outages, protecting against both data loss and hardware damage

Theft of hardware/drives Rack-mounted cabinets with cable locks, CCTV coverage, and asset tagging so removal is both harder and traceable

Task 4: Draw a simple floor plan (hand-drawn or digital) of a small office and mark at least 3 physical security measures you would implement to protect network equipment (such as CCTV, biometric access, or cable locks).



Task 5 :Use ChatGPT or a similar AI tool to generate a checklist of 7 actions a network admin should take to mitigate risks from both network threats and physical security breaches, then review and edit the checklist to make it specific for a college computer lab environment.

Segment the lab network - put lab PCs on their own VLAN, separate from staff/admin networks, so a compromised student machine can't reach sensitive college systems.

Enforce login and password policy - unique student logins (not one shared lab account), forced password changes each semester, and auto-lock after inactivity between class sessions.

Lock down endpoints - disable USB/removable media where possible, restrict software installation rights for student accounts, and keep OS + antivirus patched via centralized deployment.

Monitor traffic - run basic IDS/logging on the lab subnet to catch unusual activity (port scanning, malware beaconing) coming from lab machines.

Secure the physical space - CCTV at the lab entrance, restricted after-hours access (card/biometric), and cable locks or secured enclosures for routers/switches inside the lab (not just the main server room).

Back up lab configurations - maintain a golden image of the lab PC setup and test restoring it regularly, since labs get re-imaged often and are a common ransomware/misconfiguration target.

Run periodic security awareness sessions - brief students each semester on phishing, password hygiene, and safe browsing, since students are usually the weakest link in a shared lab environment.